# Block practice · More breathing and gas exchange

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Explain how airway features keep air clean and flowing

Use a spirometer and identify lung volumes on its trace

Calculate breathing rate or air volume from trace or table

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# Practice workbook · Explain how airway features keep air clean and flowing

Nasal warming and filtering, cartilage rings, mucus and cilia, and what happens when bronchitis or smoke damages them.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-AIRWAY-PROTECTION-AIRFLOW

# 1. Question 6 · part b · 2 marks

June 2019 Paper 1

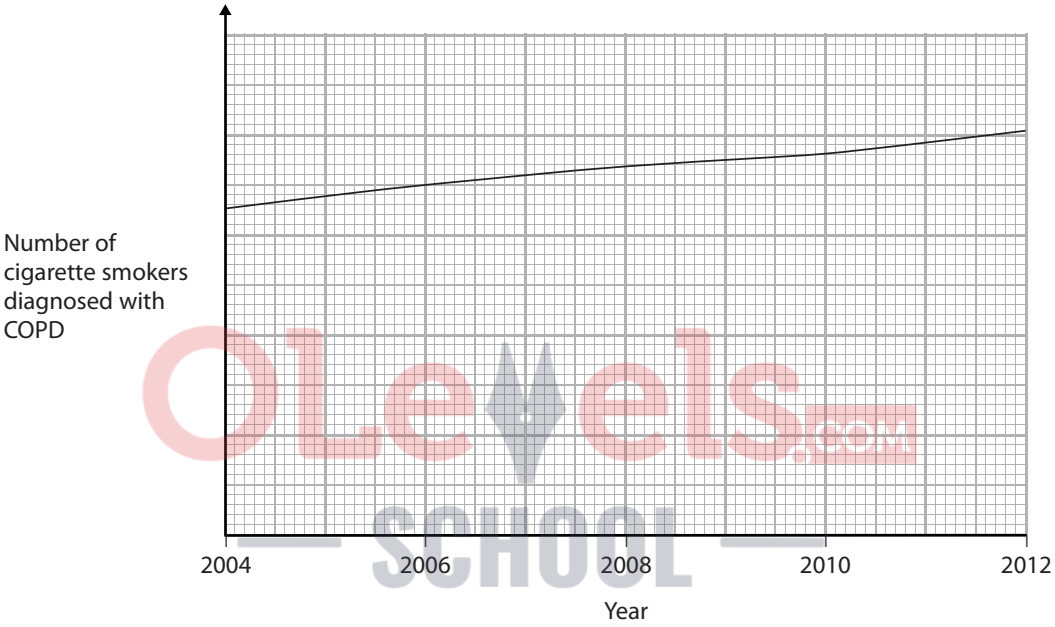
Worked-example retry · Attempt part b only.

**6** (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.




(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

**(1)**

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(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

**(2)**

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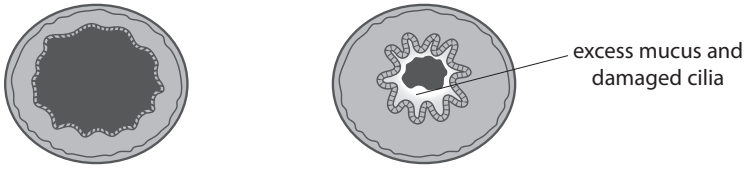
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# 1. Question 6 · part b · 2 marks

June 2019 Paper 1

Worked-example retry · Attempt part b only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.




healthy airway

chronic bronchitis

Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

**(2)**

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# 1. Question 6 · part b · 2 marks

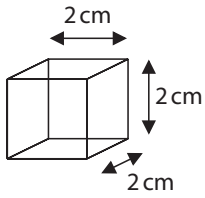
June 2019 Paper 1

Worked-example retry · Attempt part b only.

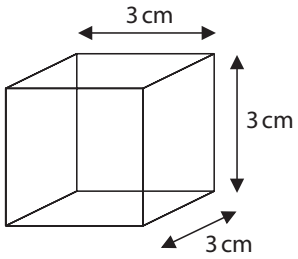
(c) The diagram shows models of two alveoli.

One model represents an alveolus from a person with healthy lungs.

The other model represents an alveolus from a person with emphysema.




**Healthy**




**Emphysema**

The table shows the surface area to volume ratio for a healthy alveolus.

| | Surface area in $cm^2$ | Volume in $cm^3$ | Surface area to volume ratio |
|---|---|---|---|
| Healthy | 24 | 8 | 3 : 1 |
| Emphysema | | | |

(i) Complete the table by giving the missing information.

**(3)**

(ii) Explain how the surface area to volume ratio of alveoli in the lungs of a person with emphysema will affect the normal function of body cells.

**(3)**

**(Total for Question 6 = 11 marks)**

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## 1. Question 6 · part b · 2 marks · Working and answer

Working and answer

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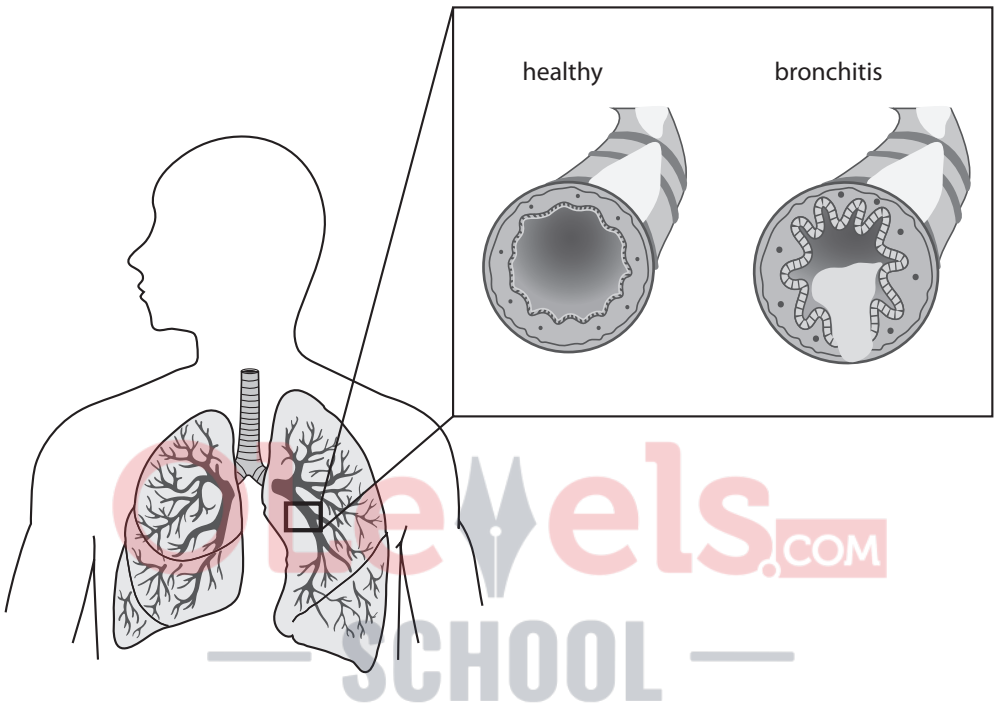
## 2. Question 6 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

**6** Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.




(a) Name the part of the lungs affected by bronchitis.

**(1)**

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## 2. Question 6 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

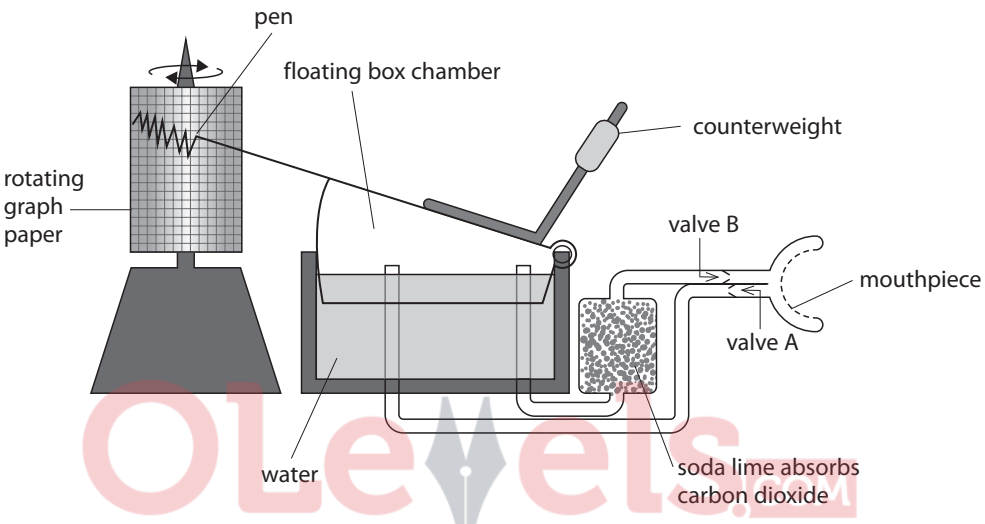
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.




(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

**(4)**

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## 2. Question 6 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

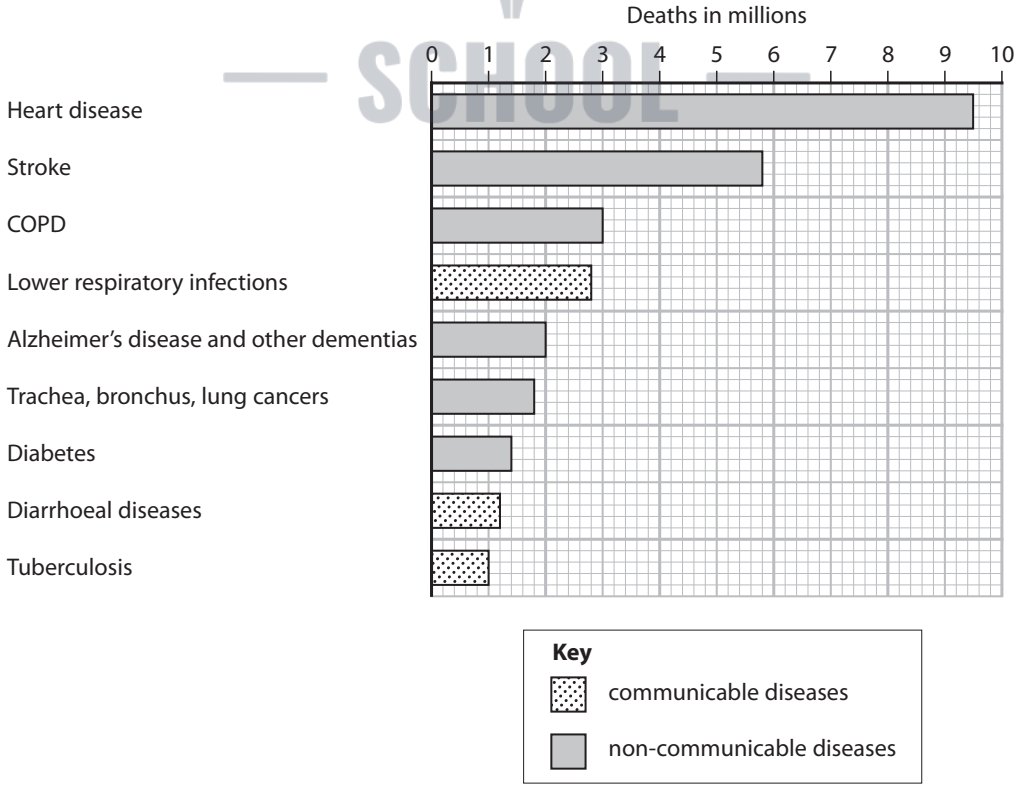
(ii) State one way to reduce risk to health when using a spirometer.

**(1)**

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

**(2)**

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.




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## 2. Question 6 · part b.iii · 2 marks · Working and answer

Working and answer

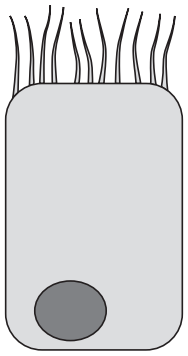
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# 3. Question 2 · part b.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

**2** The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

**(3)**

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

**(3)**

| Structure | Function |
|---|---|
| cilia | holds the genetic code |
| cell membrane | site where most chemical reactions occur |
| nucleus | controls the entry and exit of substances |
| | makes proteins |
| | moves mucus through the trachea |

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# 3. Question 2 · part b.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

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(b) (i) Cigarette smoking affects the function of the cell shown in the diagram.

Name a substance in cigarette smoke that causes damage to this cell.

**(1)**

.....

(ii) Describe how cigarette smoking affects the cell shown in the diagram.

**(2)**

.....

.....

.....

.....

**(Total for Question 2 = 9 marks)**

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Working and answer

# 4. Question 2 · part a.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

**2** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 $dm^3$ of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

**(3)**

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# 4. Question 2 · part a.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

**(2)**

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

**(1)**

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

**(1)**

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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## 4. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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## 5. Question 2 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

**2** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

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Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

**(3)**

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# 5. Question 2 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

**(2)**

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.....

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

**(1)**

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

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(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

**(1)**

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

## 5. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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# Practice workbook · Use a spirometer and identify lung volumes on its trace

How a spirometer records breathing, describing its use to measure tidal volume, and recognising tidal volume, vital capacity and residual volume or exercise changes on a trace.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SPIROMETER-TRACE-INTERPRET

# 1. Question 5 · part b.i · 4 marks

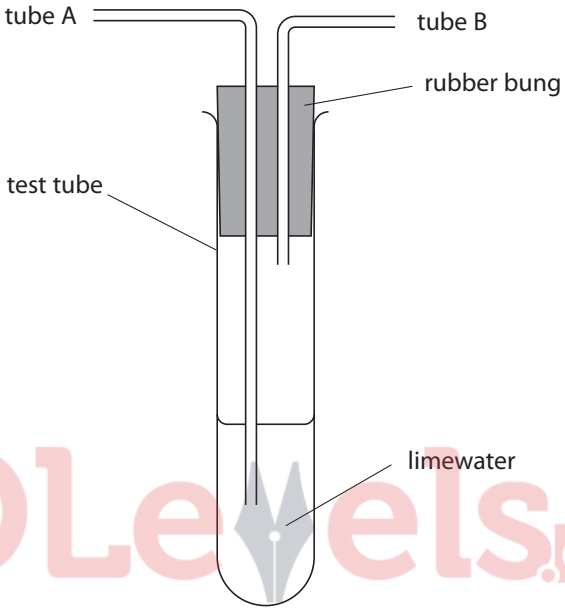
November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

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**5** The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.




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(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

**(4)**

# 1. Question 5 · part b.i · 4 marks

November 2021 Paper 1

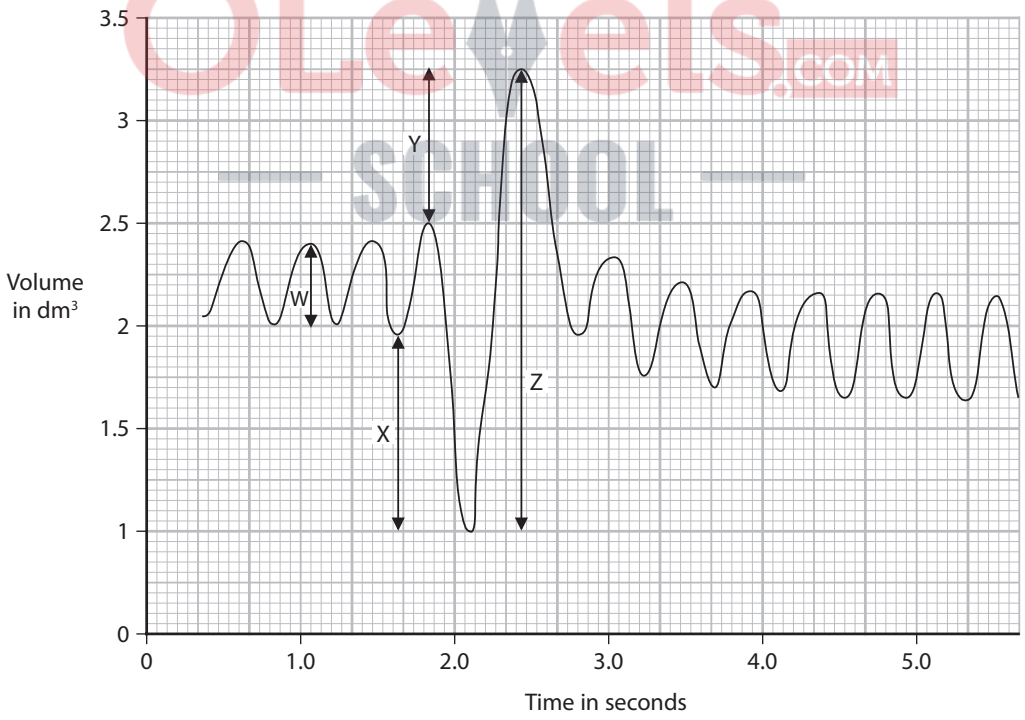
Worked-example retry · Attempt part b.i only.

(ii) Explain the difference expected between the results.

**(3)**

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.




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# 1. Question 5 · part b.i · 4 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

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(i) Complete the table using the correct letters from the trace to show the tidal volume, the vital capacity and the volume of air in each case.

**(4)**

| Lung volume | Letter | Volume in $dm^3$ |
|---|---|---|
| tidal volume | | |
| vital capacity | | |

(ii) Explain the pattern of breathing occurring to give lung volume Z.

**(2)**

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**(Total for Question 5 = 13 marks)**

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## 1. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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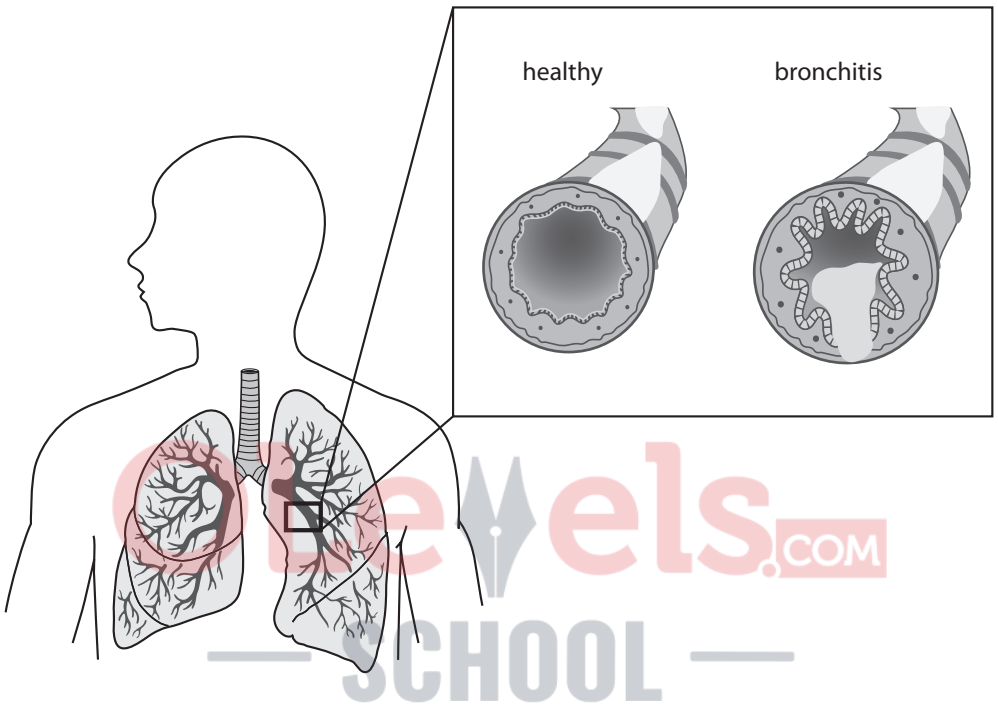
## 2. Question 6 · part b.i · 4 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

**6** Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.




(a) Name the part of the lungs affected by bronchitis.

**(1)**

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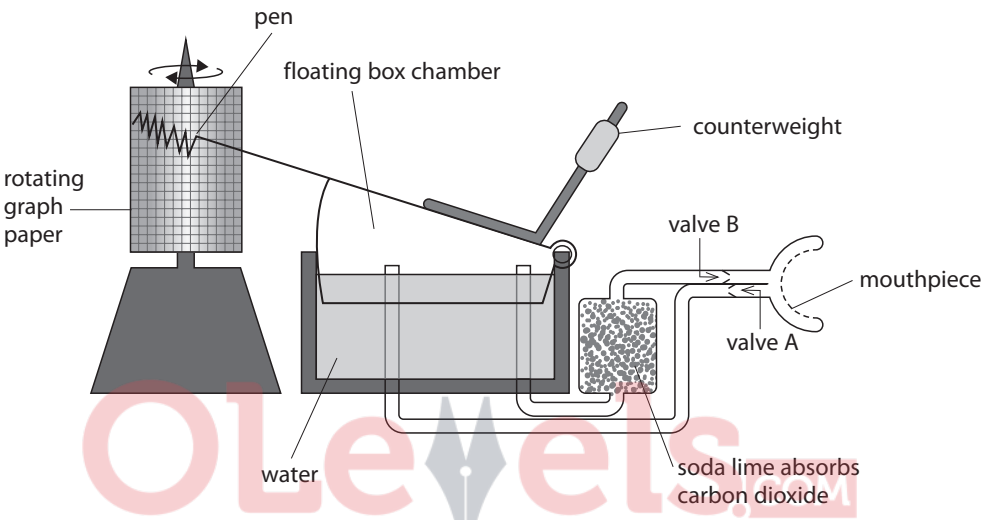
## 2. Question 6 · part b.i · 4 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.




(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

**(4)**

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## 2. Question 6 · part b.i · 4 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

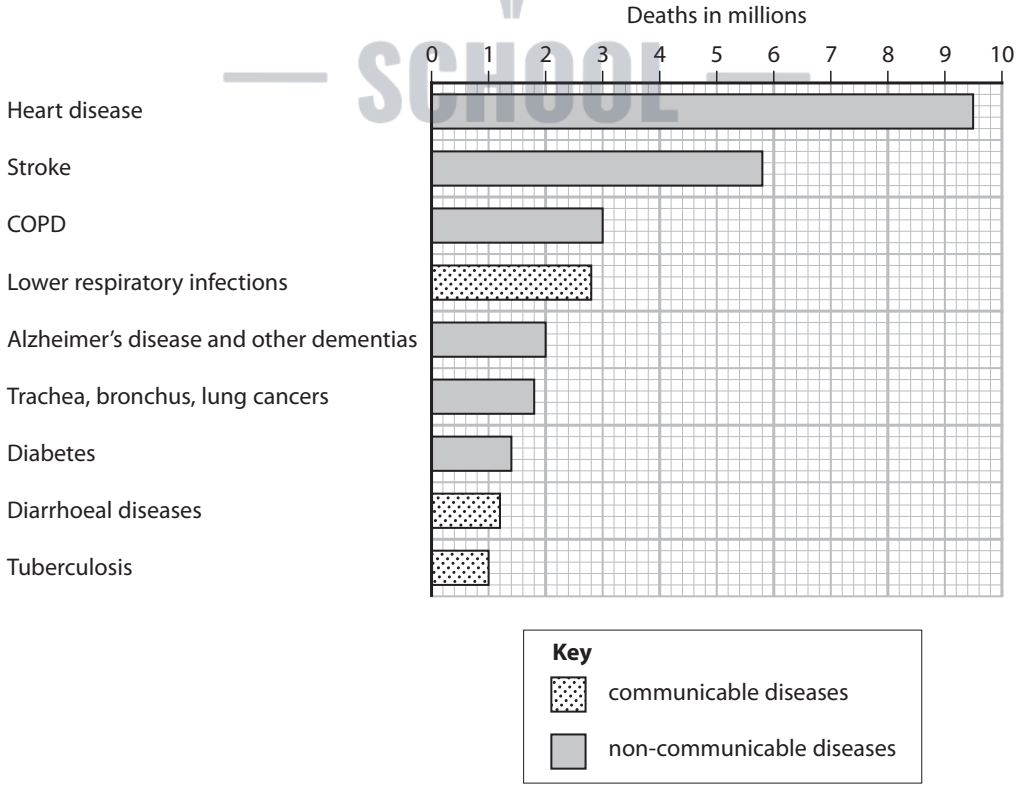
(ii) State one way to reduce risk to health when using a spirometer.

**(1)**

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

**(2)**

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.




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## 2. Question 6 · part b.i · 4 marks · Working and answer

Working and answer

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## 3. Question 2 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

**2** Aerobic respiration is the process that releases energy for use in the body.

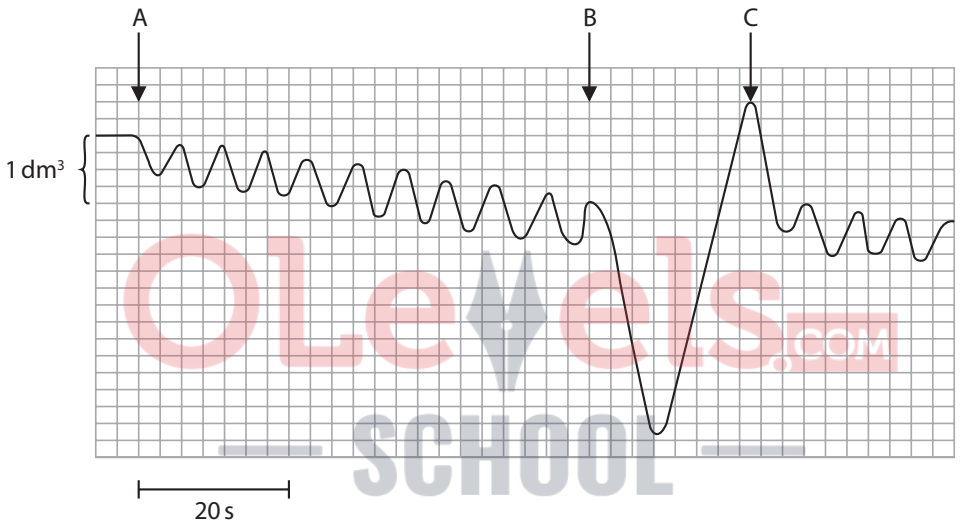
(a) Give the word equation for the process of aerobic respiration.

**(4)**

..... + ..... → ..... + ..... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.




(i) Calculate the rate of breathing between points A and B shown on the trace.

**(2)**

rate = ..... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

**(3)**

difference in volume = ..... $dm^3$

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## 3. Question 2 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

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(c) Describe the differences that would be seen in the trace between points A and B if the person had been exercising vigorously before the readings were taken.

**(2)**

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**(Total for Question 2 = 11 marks)**

Working and answer

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# 4. Question 5 · part b.ii · 2 marks

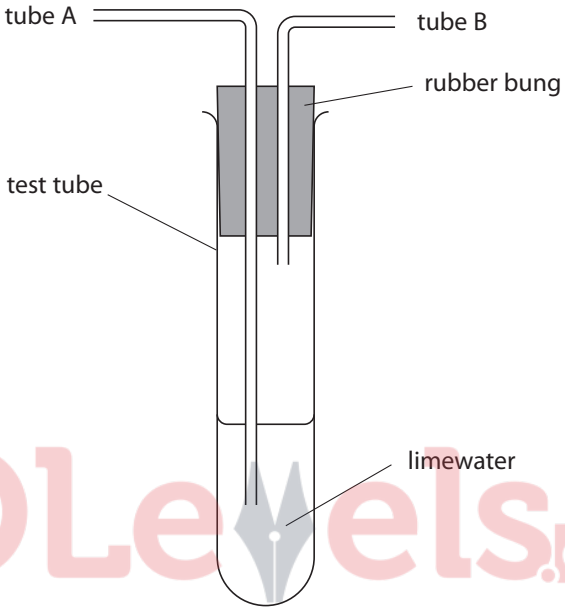
November 2021 Paper 1

Further practice · Attempt part b.ii only.

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**5** The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.




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(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

**(4)**

## 4. Question 5 · part b.ii · 2 marks

November 2021 Paper 1

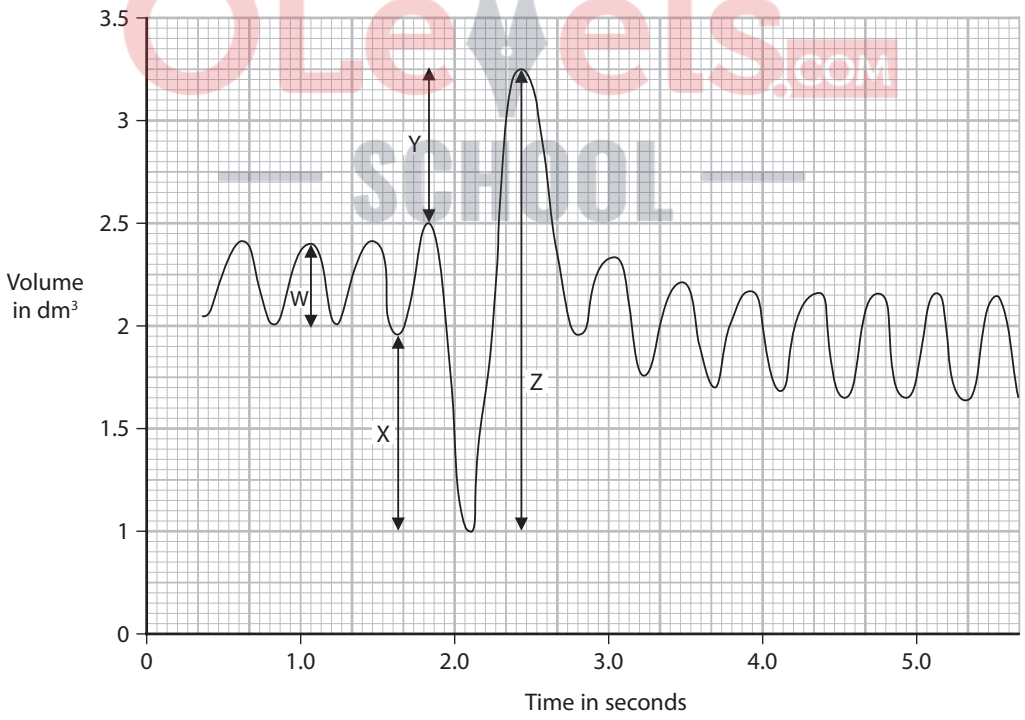
Further practice · Attempt part b.ii only.

(ii) Explain the difference expected between the results.

**(3)**

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.




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# 4. Question 5 · part b.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(i) Complete the table using the correct letters from the trace to show the tidal volume, the vital capacity and the volume of air in each case.

**(4)**

| Lung volume | Letter | Volume in $dm^3$ |
|---|---|---|
| tidal volume | | |
| vital capacity | | |

(ii) Explain the pattern of breathing occurring to give lung volume Z.

**(2)**

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**(Total for Question 5 = 13 marks)**

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Working and answer

# 5. Question 7 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

**7** (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

**(4)**

(b) Describe the role of chemoreceptors in the carotid arteries.

**(2)**

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## 5. Question 7 · part c.i · 1 mark

June 2024 Paper 1

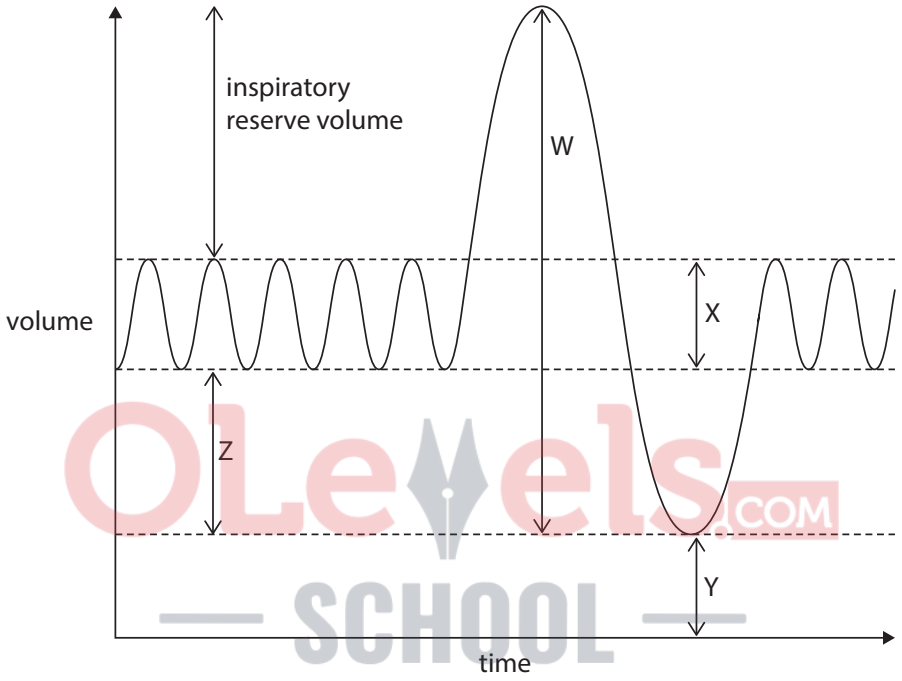
Further practice · Attempt part c.i only.

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(c) The graph shows different lung volumes.




(Source: https://www.inds.co.uk/education/how2/measure-lung-volume/)

(i) Give the name of the lung volume represented by letter X.

**(1)**

.....

(ii) Give the name of the lung volume represented by letter Y.

**(1)**

.....

(iii) Give the name of the lung volume represented by letter W.

**(1)**

.....

**(Total for Question 7 = 9 marks)**

## 5. Question 7 · part c.i · 1 mark · Working and answer

Working and answer

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## 6. Question 7 · part c.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.ii only.

**7** (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

**(4)**

(b) Describe the role of chemoreceptors in the carotid arteries.

**(2)**

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## 6. Question 7 · part c.ii · 1 mark

June 2024 Paper 1

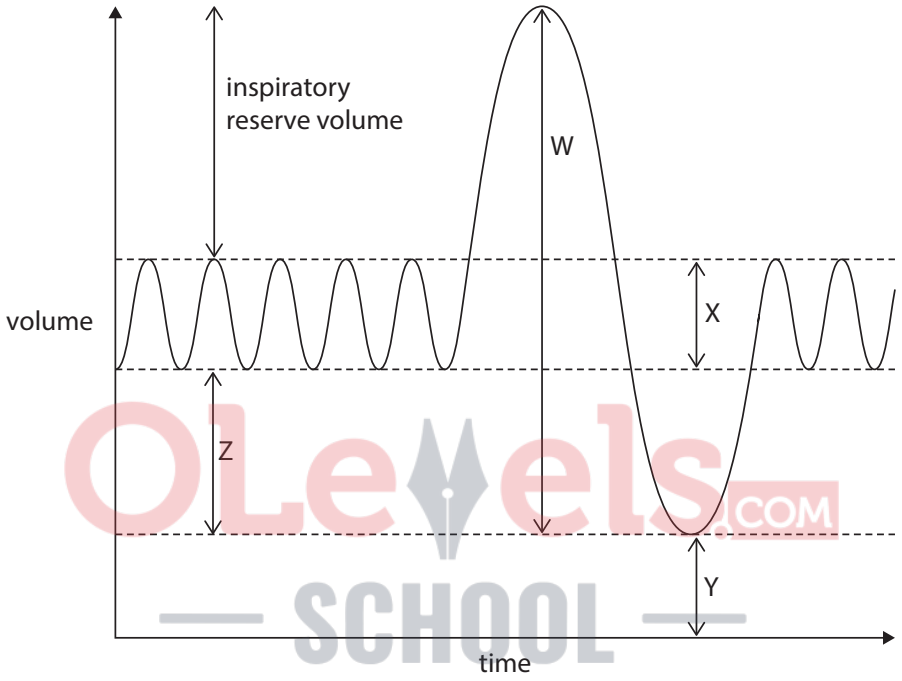
Further practice · Attempt part c.ii only.

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(c) The graph shows different lung volumes.




(Source: https://www.inds.co.uk/education/how2/measure-lung-volume/)

(i) Give the name of the lung volume represented by letter X.

**(1)**

.....

(ii) Give the name of the lung volume represented by letter Y.

**(1)**

.....

(iii) Give the name of the lung volume represented by letter W.

**(1)**

.....

**(Total for Question 7 = 9 marks)**

## 6. Question 7 · part c.ii · 1 mark · Working and answer

Working and answer

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# 7. Question 7 · part c.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.iii only.

**7** (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

**(4)**

(b) Describe the role of chemoreceptors in the carotid arteries.

**(2)**

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## 7. Question 7 · part c.iii · 1 mark

June 2024 Paper 1

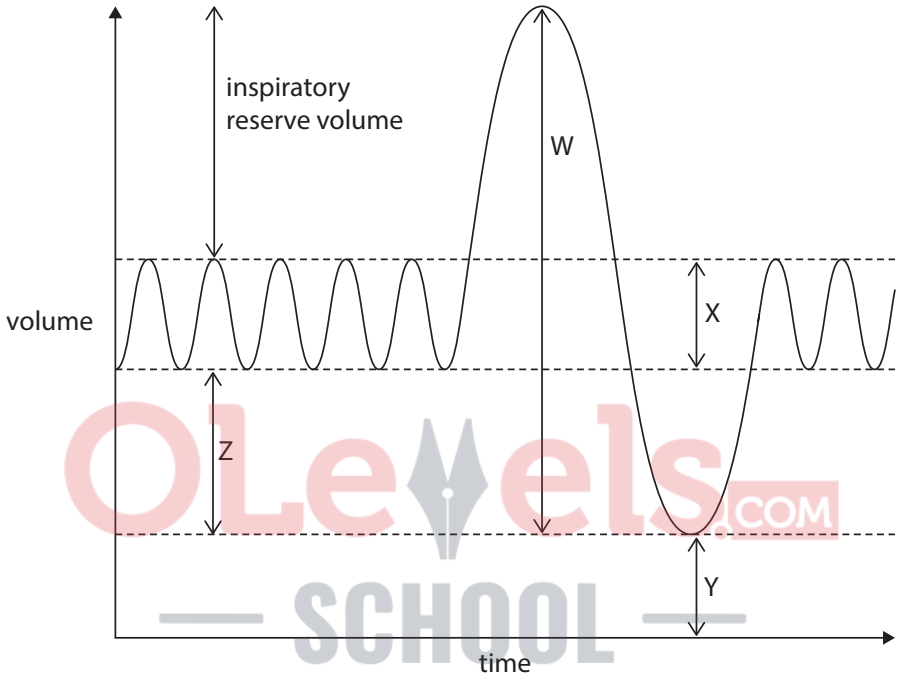
Further practice · Attempt part c.iii only.

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(i) Give the name of the lung volume represented by letter X.

**(1)**

.....

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**(1)**

.....

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**(1)**

.....

**(Total for Question 7 = 9 marks)**

## 7. Question 7 · part c.iii · 1 mark · Working and answer

Working and answer

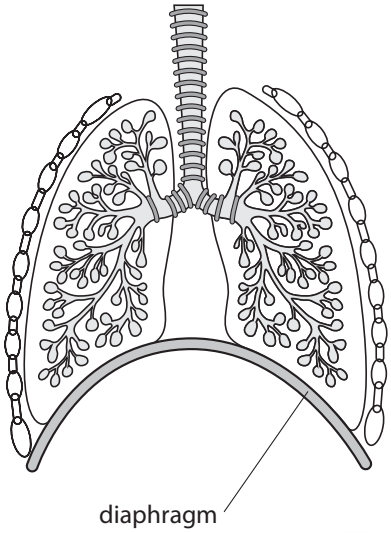
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# 8. Question 4 · part b.i · 4 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

**4** (a) The diagram shows the human lungs.




(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

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P 7 4 6 3 0 A 0 1 2 2 4

# 8. Question 4 · part b.i · 4 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

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(b) A spirometer can be used to test lung function.

(i) Describe how a spirometer is used to obtain values for tidal volume.

**(4)**

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(ii) Asthma is a condition that causes the tubes in the lungs to narrow.

A person without asthma has a vital capacity of 4600 $cm^3$.

People with severe asthma have a 30% lower vital capacity compared with a healthy person.

Calculate the vital capacity of a person with severe asthma.

**(2)**

vital capacity of a person with severe asthma = ..... $cm^3$

**(Total for Question 4 = 13 marks)**

## 8. Question 4 · part b.i · 4 marks · Working and answer

Working and answer

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# Practice workbook · Calculate breathing rate or air volume from trace or table

Numeric work on breathing: breaths per minute, volume differences, minute volume and oxygen used.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BREATHING-RATE-VOLUME-CALC

# 1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

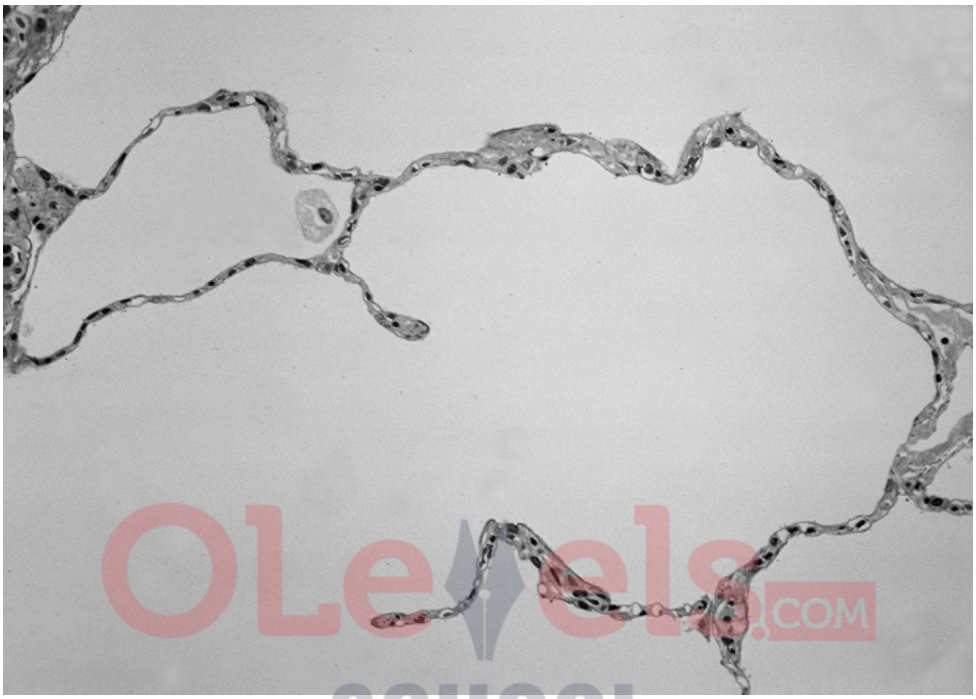
Worked-example retry · Attempt part b.iii only.

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**2** The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

**(2)**

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

**(2)**

1 .....

.....

2 .....

.....

# 1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part b.iii only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

**(1)**

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

| **Percentage (%) carbon dioxide in air** | 0.02 | 1.60 | 3.00 | 6.00 |
|---|---|---|---|---|
| **Tidal volume in $cm^3$** | 500 | 700 | 1200 | 2100 |
| **Rate of breathing in breaths per minute** | 14 | 15 | 16 | 28 |

Calculate the volume of air in $dm^3$ breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 $dm^3$ = 1000 $cm^3$]

**(2)**

volume of air = ..... $dm^3$

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

**(3)**

volume of oxygen = ..... $cm^3$

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# 1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part b.iii only.

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(iv) Explain the effect of increasing the percentage of carbon dioxide in the air a person breathes in on the tidal volume and rate of breathing.

**(4)**

**(Total for Question 2 = 14 marks)**

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Working and answer

# 2. Question 2 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

**2** Aerobic respiration is the process that releases energy for use in the body.

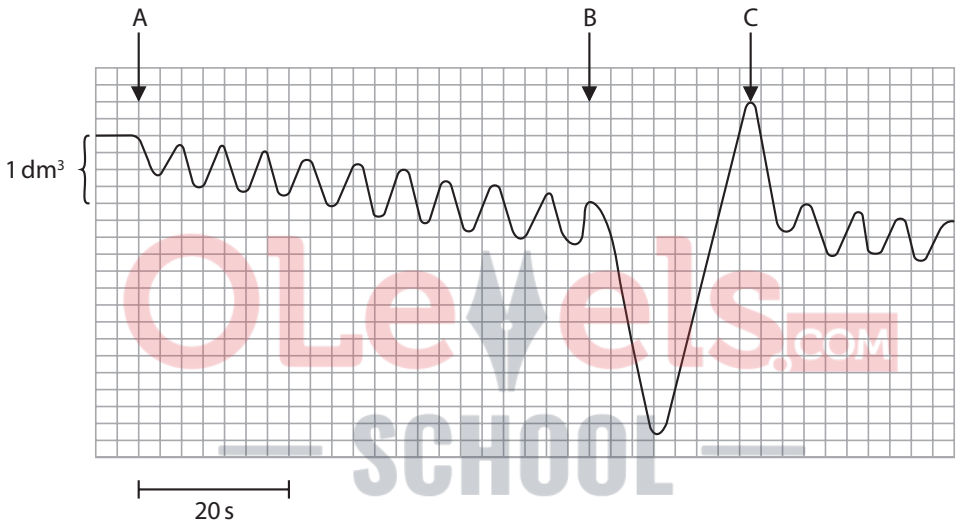
(a) Give the word equation for the process of aerobic respiration.

**(4)**

..... + ..... → ..... + ..... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.




(i) Calculate the rate of breathing between points A and B shown on the trace.

**(2)**

rate = ..... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

**(3)**

difference in volume = ..... $dm^3$

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P 6 6 4 3 3 A 0 4 2 4

## 2. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

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# 3. Question 2 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

**2** Aerobic respiration is the process that releases energy for use in the body.

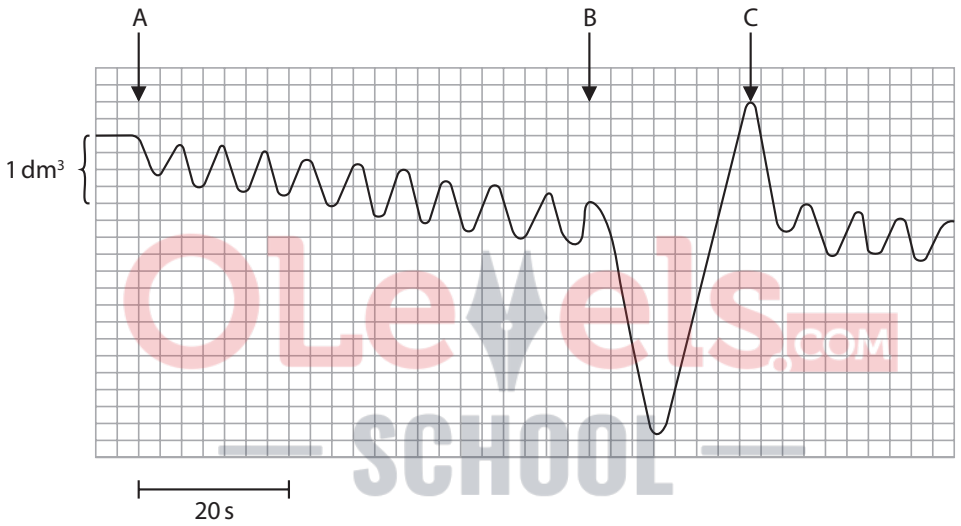
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**(2)**

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**(3)**

difference in volume = ..... $dm^3$

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## 3. Question 2 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

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(c) Describe the differences that would be seen in the trace between points A and B if the person had been exercising vigorously before the readings were taken.

**(2)**

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**(Total for Question 2 = 11 marks)**

Working and answer

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# 4. Question 2 · part b.ii · 2 marks

June 2023 Paper 1

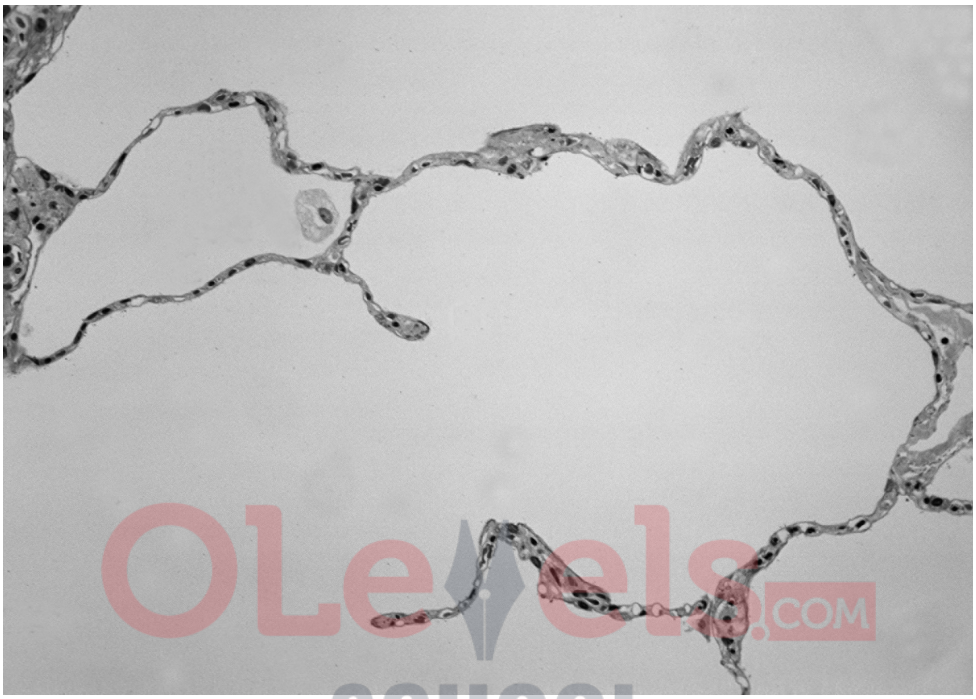
Further practice · Attempt part b.ii only.

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**2** The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

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**(2)**

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

**(2)**

1 .....

.....

2 .....

.....

# 4. Question 2 · part b.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

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DO NOT WRITE IN THIS AREA



P 7 2 5 8 5 A 0 4 2 0

## 4. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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